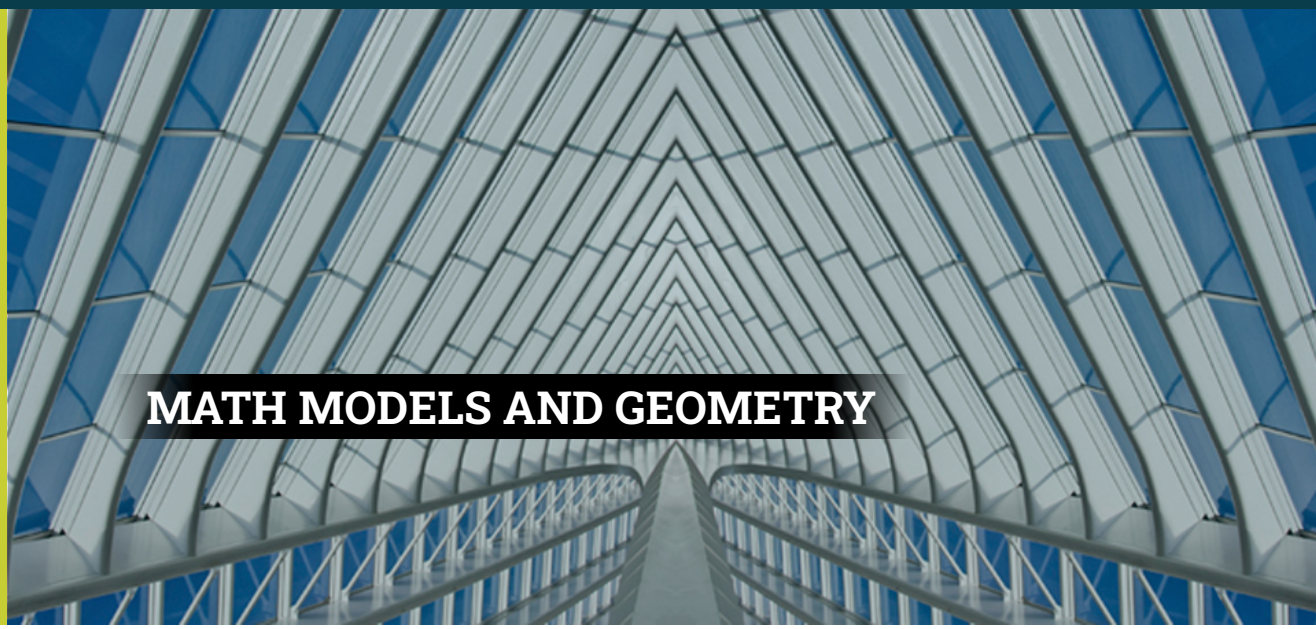


9 MATH MODELS AND GEOMETRY

Figure 9.1 Note the many individual shapes in this building. (credit: Bert Kaufmann, Flickr)

Chapter Outline

9.1 Use a Problem Solving Strategy
9.2 Solve Money Applications
9.3 Use Properties of Angles, Triangles, and the Pythagorean Theorem
9.4 Use Properties of Rectangles, Triangles, and Trapezoids
9.5 Solve Geometry Applications: Circles and Irregular Figures
9.6 Solve Geometry Applications: Volume and Surface Area
9.7 Solve a Formula for a Specific Variable

Introduction

We are surrounded by all sorts of geometry. Architects use geometry to design buildings. Artists create vivid images out of colorful geometric shapes. Street signs, automobiles, and product packaging all take advantage of geometric properties. In this chapter, we will begin by considering a formal approach to solving problems and use it to solve a variety of common problems, including making decisions about money. Then we will explore geometry and relate it to everyday situations, using the problem-solving strategy we develop.

9.1 Use a Problem Solving Strategy

Learning Objectives

By the end of this section, you will be able to:

- Approach word problems with a positive attitude
- Use a problem solving strategy for word problems
- Solve number problems

BE PREPARED 9.1 Before you get started, take this readiness quiz.

Translate "6 less than twice x" into an algebraic expression.
If you missed this problem, review Example 2.25.

BE PREPARED 9.2 Solve: $\frac{2}{3}x = 24$.
If you missed this problem, review Example 8.16.

BE PREPARED 9.3 Solve: $3x + 8 = 14$.

If you missed this problem, review Example 8.20.

Approach Word Problems with a Positive Attitude

The world is full of word problems. How much money do I need to fill the car with gas? How much should I tip the server at a restaurant? How many socks should I pack for vacation? How big a turkey do I need to buy for Thanksgiving dinner, and what time do I need to put it in the oven? If my sister and I buy our mother a present, how much will each of us pay?

Now that we can solve equations, we are ready to apply our new skills to word problems. Do you know anyone who has had negative experiences in the past with word problems? Have you ever had thoughts like the student in Figure 9.2?

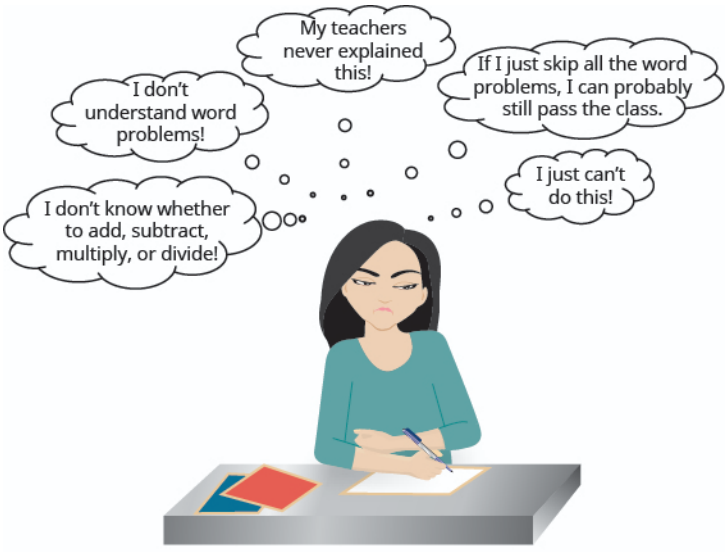


Figure 9.2 Negative thoughts about word problems can be barriers to success.

When we feel we have no control, and continue repeating negative thoughts, we set up barriers to success. We need to calm our fears and change our negative feelings.

Start with a fresh slate and begin to think positive thoughts like the student in Figure 9.3. Read the positive thoughts and say them out loud.



Figure 9.3 When it comes to word problems, a positive attitude is a big step toward success.

If we take control and believe we can be successful, we will be able to master word problems.

Think of something that you can do now but couldn't do three years ago. Whether it's driving a car, snowboarding, cooking a gourmet meal, or speaking a new language, you have been able to learn and master a new skill. Word problems are no different. Even if you have struggled with word problems in the past, you have acquired many new math skills that will help you succeed now!

Use a Problem-solving Strategy for Word Problems

In earlier chapters, you translated word phrases into algebraic expressions, using some basic mathematical vocabulary and symbols. Since then you've increased your math vocabulary as you learned about more algebraic procedures, and you've had more practice translating from words into algebra.

You have also translated word sentences into algebraic equations and solved some word problems. The word problems applied math to everyday situations. You had to restate the situation in one sentence, assign a variable, and then write an equation to solve. This method works as long as the situation is familiar to you and the math is not too complicated.

Now we'll develop a strategy you can use to solve any word problem. This strategy will help you become successful with word problems. We'll demonstrate the strategy as we solve the following problem.

EXAMPLE 9.1

Pete bought a shirt on sale for $\$18$, which is one-half the original price. What was the original price of the shirt?

Solution

Step 1. **Read** the problem. Make sure you understand all the words and ideas. You may need to read the problem two or more times. If there are words you don't understand, look them up in a dictionary or on the Internet.

- *In this problem, do you understand what is being discussed? Do you understand every word?*

Step 2. **Identify** what you are looking for. It's hard to find something if you are not sure what it is! Read the problem again and look for words that tell you what you are looking for!

- *In this problem, the words "what was the original price of the shirt" tell you that what you are looking for: the original price of the shirt.*

Step 3. **Name** what you are looking for. Choose a variable to represent that quantity. You can use any letter for the variable, but it may help to choose one that helps you remember what it represents.

- *Let p = the original price of the shirt*

Step 4. **Translate** into an equation. It may help to first restate the problem in one sentence, with all the important information. Then translate the sentence into an equation.

18	is	one-half	of	the original price .
18	=	$\frac{1}{2}$	·	p

Step 5. **Solve** the equation using good algebra techniques. Even if you know the answer right away, using algebra will better prepare you to solve problems that do not have obvious answers.

Write the equation.	$18 = \frac{1}{2}p$
Multiply both sides by 2.	$2 \cdot 18 = 2 \cdot \frac{1}{2}p$
Simplify.	$36 = p$

Step 6. **Check** the answer in the problem and make sure it makes sense.

- *We found that $p = 36$, which means the original price was $\$36$. Does $\$36$ make sense in the problem? Yes, because 18 is one-half of 36, and the shirt was on sale at half the original price.*

Step 7. **Answer** the question with a complete sentence.

- *The problem asked "What was the original price of the shirt?" The answer to the question is: "The original price of the shirt was* $\$36$ *."*

If this were a homework exercise, our work might look like this:

Let p = the original price.

18 is one-half the original price.

$$18 = \frac{1}{2}p$$

$$2 \cdot 18 = 2 \cdot \frac{1}{2}p$$

$$36 = p$$

Check:

Is \$36 a reasonable price for a shirt? *Yes*.

Is 18 one-half of 36? *Yes*.

The original price of the shirt was \$36.

TRY IT 9.1 Joaquin bought a bookcase on sale for $\$120$, which was two-thirds the original price. What was the original price of the bookcase?

TRY IT 9.2 Two-fifths of the people in the senior center dining room are men. If there are 16 men, what is the total number of people in the dining room?

We list the steps we took to solve the previous example.

Problem-Solving Strategy

Step 1. **Read** the word problem. Make sure you understand all the words and ideas. You may need to read the problem two or more times. If there are words you don't understand, look them up in a dictionary or on the internet.
Step 2. **Identify** what you are looking for.
Step 3. **Name** what you are looking for. Choose a variable to represent that quantity.
Step 4. **Translate** into an equation. It may be helpful to first restate the problem in one sentence before translating.
Step 5. **Solve** the equation using good algebra techniques.
Step 6. **Check** the answer in the problem. Make sure it makes sense.
Step 7. **Answer** the question with a complete sentence.

Let's use this approach with another example.

EXAMPLE 9.2

Yash brought apples and bananas to a picnic. The number of apples was three more than twice the number of bananas. Yash brought 11 apples to the picnic. How many bananas did he bring?

Solution

Step 1. **Read** the problem.	
Step 2. **Identify** what you are looking for.	How many bananas did he bring?

Step 3. **Name** what you are looking for. Choose a variable to represent the number of bananas.	Let b = number of bananas
Step 4. **Translate.** Restate the problem in one sentence with all the important information. Translate into an equation.	The number of apples was 3 more than twice the number of bananas. $11 = 3 + 2b$
Step 5. **Solve** the equation.	$11 = 2b + 3$
Subtract 3 from each side.	$11 - 3 = 2b + 3 - 3$
Simplify.	$8 = 2b$
Divide each side by 2.	$\frac{8}{2} = \frac{2b}{2}$
Simplify.	$4 = b$
Step 6. **Check:** First, is our answer reasonable? Yes, bringing four bananas to a picnic seems reasonable. The problem says the number of apples was three more than twice the number of bananas. If there are four bananas, does that make eleven apples? Twice 4 bananas is 8. Three more than 8 is 11.	
Step 7. **Answer** the question.	Yash brought 4 bananas to the picnic.

TRY IT 9.3 Guillermo bought textbooks and notebooks at the bookstore. The number of textbooks was 3 more than the number of notebooks. He bought 5 textbooks. How many notebooks did he buy?

TRY IT 9.4 Gerry worked Sudoku puzzles and crossword puzzles this week. The number of Sudoku puzzles he completed is seven more than the number of crossword puzzles. He completed 14 Sudoku puzzles. How many crossword puzzles did he complete?

In Solve Sales Tax, Commission, and Discount Applications, we learned how to translate and solve basic percent equations and used them to solve sales tax and commission applications. In the next example, we will apply our Problem Solving Strategy to more applications of percent.

EXAMPLE 9.3

Nga's car insurance premium increased by $60, which was 8% of the original cost. What was the original cost of the premium?

Solution

Step 1. **Read** the problem. Remember, if there are words you don't understand, look them up.	
Step 2. **Identify** what you are looking for.	the original cost of the premium
Step 3. **Name.** Choose a variable to represent the original cost of premium.	Let $c =$ the original cost
Step 4. **Translate.** Restate as one sentence. Translate into an equation.	$\underbrace{\$60}_{60}\ \underbrace{\text{was}}_{=}\ \underbrace{8\%}_{0.08}\ \underbrace{\text{of}}_{\cdot}\ \underbrace{\text{the original cost}}_{c}.$
Step 5. **Solve** the equation.	$60 = 0.08c$
Divide both sides by 0.08.	$\frac{60}{0.08} = \frac{0.08c}{0.08}$
Simplify.	$c = 750$
Step 6. **Check:** Is our answer reasonable? Yes, a \$750 premium on auto insurance is reasonable. Now let's check our algebra. Is 8% of 750 equal to 60? $750 = c$ $0.08(750) = 60$ $60 = 60$ ✓	
Step 7. **Answer** the question.	The original cost of Nga's premium was \$750.

TRY IT 9.5 Pilar's rent increased by 4%. The increase was $\$38$. What was the original amount of Pilar's rent?

TRY IT 9.6 Steve saves 12% of his paycheck each month. If he saved $\$504$ last month, how much was his paycheck?

Solve Number Problems

Now we will translate and solve **number problems**. In number problems, you are given some clues about one or more numbers, and you use these clues to build an equation. Number problems don't usually arise on an everyday basis, but they provide a good introduction to practicing the Problem Solving Strategy. Remember to look for clue words such as *difference*, *of*, and *and*.

EXAMPLE 9.4

The difference of a number and six is 13. Find the number.

Solution

Step 1. **Read** the problem. Do you understand all the words?	
Step 2. **Identify** what you are looking for.	the number
Step 3. **Name.** Choose a variable to represent the number.	Let $n =$ the number
Step 4. **Translate.** Restate as one sentence. Translate into an equation.	$\underbrace{\text{The difference of a number and 6}}\ \underbrace{\text{is}}\ \underbrace{\text{13.}}$ $n - 6 \quad = \quad 13$
Step 5. **Solve** the equation. Add 6 to both sides. Simplify.	$n - 6 = 13$ $n - 6 + 6 = 13 + 6$ $n = 19$
Step 6. **Check:** The difference of 19 and 6 is 13. It checks.	
Step 7. **Answer** the question.	The number is 19.

TRY IT 9.7 The difference of a number and eight is 17. Find the number.

TRY IT 9.8 The difference of a number and eleven is −7. Find the number.

EXAMPLE 9.5

The sum of twice a number and seven is 15. Find the number.

Solution

Step 1. **Read** the problem.	
Step 2. **Identify** what you are looking for.	the number
Step 3. **Name.** Choose a variable to represent the number.	Let $n =$ the number
Step 4. **Translate.** Restate the problem as one sentence. Translate into an equation.	$\underbrace{\text{The sum of twice a number}}\ \underbrace{\text{and}}\ \underbrace{\text{7}}\ \underbrace{\text{is}}\ \underbrace{\text{15.}}$ $2 \cdot n \quad + \quad 7 \quad = \quad 15$
Step 5. **Solve** the equation.	$2n + 7 = 15$
Subtract 7 from each side and simplify.	$2n = 8$
Divide each side by 2 and simplify.	$n = 4$

Step 6. **Check:** is the sum of twice 4 and 7 equal to 15?	$2 \cdot 4 + 7 = 15$ $8 + 7 = 15$ $15 = 15$ ✓
Step 7. **Answer** the question.	The number is 4.

TRY IT 9.9 The sum of four times a number and two is 14. Find the number.

TRY IT 9.10 The sum of three times a number and seven is 25. Find the number.

Some number word problems ask you to find two or more numbers. It may be tempting to name them all with different variables, but so far we have only solved equations with one variable. We will define the numbers in terms of the same variable. Be sure to read the problem carefully to discover how all the numbers relate to each other.

EXAMPLE 9.6

One number is five more than another. The sum of the numbers is twenty-one. Find the numbers.

Solution

Step 1. **Read** the problem.		
Step 2. **Identify** what you are looking for.		You are looking for two numbers.
Step 3. **Name.** Choose a variable to represent the first number. What do you know about the second number? Translate.		Let $n =$ 1st number One number is five more than another. $n + 5 =$ 2nd number
Step 4. **Translate.** Restate the problem as one sentence with all the important information. Translate into an equation. Substitute the variable expressions.		The sum of the numbers is 21. The sum of the 1st number and the 2nd number is 21. $\underbrace{\text{1st number}}_{n} + \underbrace{\text{2nd number}}_{n+5} = 21$ $n + n + 5 = 21$
Step 5. **Solve** the equation.		$n + n + 5 = 21$
Combine like terms.		$2n + 5 = 21$
Subtract five from both sides and simplify.		$2n = 16$
Divide by two and simplify.		$n = 8$ 1st number
Find the second number too.		$n + 5$ 2nd number
Substitute $n = 8$		$8 + 5$

		13
Step 6. **Check:**		
Do these numbers check in the problem? Is one number 5 more than the other? Is thirteen, 5 more than 8? Yes. Is the sum of the two numbers 21?	$13 \stackrel{?}{=} 8+5$ $13=13$ ✓ $8+13 \stackrel{?}{=} 21$ $21=21$ ✓	
Step 7. **Answer** the question.		The numbers are 8 and 13.

TRY IT 9.11 One number is six more than another. The sum of the numbers is twenty-four. Find the numbers.

TRY IT 9.12 The sum of two numbers is fifty-eight. One number is four more than the other. Find the numbers.

EXAMPLE 9.7

The sum of two numbers is negative fourteen. One number is four less than the other. Find the numbers.

Solution

Step 1. **Read** the problem.		
Step 2. **Identify** what you are looking for.		two numbers
Step 3. **Name.** Choose a variable. What do you know about the second number? Translate.		Let n = 1st number One number is 4 less than the other. n - 4 = 2nd number
Step 4. **Translate.** Write as one sentence. Translate into an equation. Substitute the variable expressions.		The sum of two numbers is negative fourteen. 1st number + 2nd number = −14 $n + n-4 = -14$
Step 5. **Solve** the equation.		$n+n-4=-14$
Combine like terms.		$2n-4=-14$
Add 4 to each side and simplify.		$2n=-10$
Divide by 2.		$n=-5$ 1st number
Substitute $n=-5$ to find the 2nd number.		$n-4$ 2nd number

		$-5-4$
		-9
Step 6. **Check:**		
Is −9 four less than −5?	$-5-4\stackrel{?}{=}-9$ $-9=-9$ ✓	
Is their sum −14?	$-5+(-9)\stackrel{?}{=}-14$ $-14=-14$ ✓	
Step 7. **Answer** the question.		The numbers are −5 and −9.

TRY IT 9.13 The sum of two numbers is negative twenty-three. One number is 7 less than the other. Find the numbers.

TRY IT 9.14 The sum of two numbers is negative eighteen. One number is 40 more than the other. Find the numbers.

EXAMPLE 9.8

One number is ten more than twice another. Their sum is one. Find the numbers.

Solution

Step 1. **Read** the problem.		
Step 2. **Identify** what you are looking for.		two numbers
Step 3. **Name.** Choose a variable. One number is ten more than twice another.		Let x = 1st number $2x+10$ = 2nd number
Step 4. **Translate.** Restate as one sentence.		Their sum is one.
Translate into an equation		The sum of the two numbers is 1. $x+(2x+10) = 1$
Step 5. **Solve** the equation.		$x+2x+10=1$
Combine like terms.		$3x+10=1$
Subtract 10 from each side.		$3x=-9$
Divide each side by 3 to get the first number.		$x=-3$
Substitute to get the second number.		$2x+10$

		$2(-3)+10$
		4
Step 6. **Check.**		
Is 4 ten more than twice −3?		$2(-3)+10 \stackrel{?}{=} 4$
Is their sum 1?		$-6+10=4$ $4=4$ ✓ $-3+4 \stackrel{?}{=} 1$ $1=1$ ✓
Step 7. **Answer** the question.		The numbers are −3 and 4.

TRY IT 9.15 One number is eight more than twice another. Their sum is negative four. Find the numbers.

TRY IT 9.16 One number is three more than three times another. Their sum is negative five. Find the numbers.

Consecutive integers are integers that immediately follow each other. Some examples of consecutive integers are:

$$...1, 2, 3, 4,...$$
$$...-10, -9, -8, -7,...$$
$$...150, 151, 152, 153,...$$

Notice that each number is one more than the number preceding it. So if we define the first integer as n, the next consecutive integer is $n+1$. The one after that is one more than $n+1$, so it is $n+1+1$, or $n+2$.

n	1st integer
$n+1$	2nd consecutive integer
$n+2$	3rd consecutive integer

EXAMPLE 9.9

The sum of two consecutive integers is 47. Find the numbers.

Solution

Step 1. **Read** the problem.		
Step 2. **Identify** what you are looking for.		two consecutive integers
Step 3. **Name.**		Let n = 1st integer n + 1 = next consecutive integer

Step 4. **Translate.** Restate as one sentence. Translate into an equation.		The sum of the integers is 47. $n+n+1 = 47$
Step 5. **Solve** the equation.		$n+n+1=47$
Combine like terms.		$2n+1=47$
Subtract 1 from each side.		$2n=46$
Divide each side by 2.		$n=23$ 1st integer
Substitute to get the second number.		$n+1$ 2nd integer
		$23+1$
		24
Step 6. **Check:**	$23+24 \stackrel{?}{=} 47$ $47=47$ ✓	
Step 7. **Answer** the question.		The two consecutive integers are 23 and 24.

TRY IT 9.17 The sum of two consecutive integers is 95. Find the numbers.

TRY IT 9.18 The sum of two consecutive integers is −31. Find the numbers.

EXAMPLE 9.10

Find three consecutive integers whose sum is 42.

Solution

Step 1. **Read** the problem.		
Step 2. **Identify** what you are looking for.		three consecutive integers
Step 3. **Name.**		Let n = 1st integer $n+1$ = 2nd consecutive integer $n+2$ = 3rd consecutive integer
Step 4. **Translate.** Restate as one sentence. Translate into an equation.		The sum of the three integers is 42. $n + n+1 + n+2 = 42$
Step 5. **Solve** the equation.		$n+n+1+n+2=42$

Combine like terms.		$3n+3=42$	
Subtract 3 from each side.		$3n=39$	
Divide each side by 3.		$n=13$	1st integer
Substitute to get the second number.		$n+1$	2nd integer
		$13+1$	
		14	
Substitute to get the third number.		$n+2$	3rd integer
		$13+2$	
		15	
Step 6. **Check:**	$13+14+15\stackrel{?}{=}42$ $42=42$ ✓		
Step 7. **Answer** the question.		The three consecutive integers are 13, 14, and 15.	

TRY IT 9.19 Find three consecutive integers whose sum is 96.

TRY IT 9.20 Find three consecutive integers whose sum is −36.

LINKS TO LITERACY

The Links to Literacy activities *Math Curse*, *Missing Mittens* and *Among the Odds and Evens* will provide you with another view of the topics covered in this section.

SECTION 9.1 EXERCISES

Practice Makes Perfect

Use a Problem-solving Strategy for Word Problems

In the following exercises, use the problem-solving strategy for word problems to solve. Answer in complete sentences.

1. Two-thirds of the children in the fourth-grade class are girls. If there are 20 girls, what is the total number of children in the class?

2. Three-fifths of the members of the school choir are women. If there are 24 women, what is the total number of choir members?

3. Zachary has 25 country music CDs, which is one-fifth of his CD collection. How many CDs does Zachary have?

4. One-fourth of the candies in a bag of are red. If there are 23 red candies, how many candies are in the bag?

5. There are 16 girls in a school club. The number of girls is 4 more than twice the number of boys. Find the number of boys in the club.

6. There are 18 Cub Scouts in Troop 645. The number of scouts is 3 more than five times the number of adult leaders. Find the number of adult leaders.

7. Lee is emptying dishes and glasses from the dishwasher. The number of dishes is 8 less than the number of glasses. If there are 9 dishes, what is the number of glasses?

8. The number of puppies in the pet store window is twelve less than the number of dogs in the store. If there are 6 puppies in the window, what is the number of dogs in the store?

9. After 3 months on a diet, Lisa had lost 12% of her original weight. She lost 21 pounds. What was Lisa's original weight?

10. Tricia got a 6% raise on her weekly salary. The raise was $\$30$ per week. What was her original weekly salary?

11. Tim left a $\$9$ tip for a $\$50$ restaurant bill. What percent tip did he leave?

12. Rashid left a $\$15$ tip for a $\$75$ restaurant bill. What percent tip did he leave?

13. Yuki bought a dress on sale for $\$72$. The sale price was 60% of the original price. What was the original price of the dress?

14. Kim bought a pair of shoes on sale for $\$40.50$. The sale price was 45% of the original price. What was the original price of the shoes?

Solve Number Problems

In the following exercises, solve each number word problem.

15. The sum of a number and eight is 12. Find the number.

16. The sum of a number and nine is 17. Find the number.

17. The difference of a number and twelve is 3. Find the number.

18. The difference of a number and eight is 4. Find the number.

19. The sum of three times a number and eight is 23. Find the number.

20. The sum of twice a number and six is 14. Find the number.

21. The difference of twice a number and seven is 17. Find the number.

22. The difference of four times a number and seven is 21. Find the number.

23. Three times the sum of a number and nine is 12. Find the number.

24. Six times the sum of a number and eight is 30. Find the number.

25. One number is six more than the other. Their sum is forty-two. Find the numbers.

26. One number is five more than the other. Their sum is thirty-three. Find the numbers.

27. The sum of two numbers is twenty. One number is four less than the other. Find the numbers.

28. The sum of two numbers is twenty-seven. One number is seven less than the other. Find the numbers.

29. A number is one more than twice another number. Their sum is negative five. Find the numbers.

30. One number is six more than five times another. Their sum is six. Find the numbers.

31. The sum of two numbers is fourteen. One number is two less than three times the other. Find the numbers.

32. The sum of two numbers is zero. One number is nine less than twice the other. Find the numbers.

33. One number is fourteen less than another. If their sum is increased by seven, the result is 85. Find the numbers.

34. One number is eleven less than another. If their sum is increased by eight, the result is 71. Find the numbers.

35. The sum of two consecutive integers is 77. Find the integers.

36. The sum of two consecutive integers is 89. Find the integers.

37. The sum of two consecutive integers is -23. Find the integers.

38. The sum of two consecutive integers is -37. Find the integers.

39. The sum of three consecutive integers is 78. Find the integers.

40. The sum of three consecutive integers is 60. Find the integers.

41. Find three consecutive integers whose sum is -36.

42. Find three consecutive integers whose sum is -3.

Everyday Math

43. **Shopping** Patty paid $\$35$ for a purse on sale for $\$10$ off the original price. What was the original price of the purse?

44. **Shopping** Travis bought a pair of boots on sale for $\$25$ off the original price. He paid $\$60$ for the boots. What was the original price of the boots?

45. **Shopping** Minh spent $\$6.25$ on 5 sticker books to give his nephews. Find the cost of each sticker book.

46. **Shopping** Alicia bought a package of 8 peaches for $\$3.20$. Find the cost of each peach.

47. **Shopping** Tom paid $\$1{,}166.40$ for a new refrigerator, including $\$86.40$ tax. What was the price of the refrigerator before tax?

48. **Shopping** Kenji paid $\$2{,}279$ for a new living room set, including $\$129$ tax. What was the price of the living room set before tax?

Writing Exercises

49. Write a few sentences about your thoughts and opinions of word problems. Are these thoughts positive, negative, or neutral? If they are negative, how might you change your way of thinking in order to do better?

50. When you start to solve a word problem, how do you decide what to let the variable represent?

Self Check

After completing the exercises, use this checklist to evaluate your mastery of the objectives of this section.

I can...	Confidently	With some help	No-I don't get it!
approach word problems with a positive attitude.			
use a problem solving strategy for word problems.			
solve number problems.			